

- Q.6 Mass transfer in food processing (e.g., drying) involves the movement of:
- Electrons
 - Moisture or solutes
 - Photons
 - Neutrons
- Q.7 The internal resistance of a fluid to flow is called:
- Density
 - Surface tension
 - Viscosity
 - Buoyancy
- Q.8 The fastest mode of heat transfer (travels at the speed of light) is:
- Conduction
 - Convection
 - Radiation
 - Osmosis
- Q.9 An autoclave uses _____ to achieve sterilization.
- Saturated steam under pressure
 - Ultraviolet light
 - Microwaves
 - Dry heat
- Q.10 scaling (fouling) in heat ex-changers leads to :
- Increased efficiency
 - Lower pressure drop
 - Faster flow
 - Decreased heat transfer rate

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The internal resistance of a fluid to flow is called _____.
- Q.12 _____ Law governs the process of heat conduction.
- Q.13 A _____ is used to measure the pressure difference in a pipe.
- Q.14 Bulk density is always higher than particle density. (True / False)

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- Q.15 Enthalpy includes both internal energy and flow work. (True / False)
- Q.16 Specific gravity has no units. (True / False)
- Q.17 The temperature at which condensation begins is the _____ point,
- Q.18 _____ is the study of deformation and flow of food materials.
- Q.19 Plate heat exchangers are commonly used for high-viscosity pastes. (True / False)
- Q.20 Bernoulli's equation is based on the conservation of mass. (True / False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the importance of thousand grain weight in the quality assessment of cereals.
- Q.22 Define Porosity. Provide the mathematical relationship between porosity, bulk density and true density.
- Q.23 Explain Thermal Conductivity and list three factors that affects its value in fresh vegetables.
- Q.24 Define Specific Heat and explain how the Siebel's equation is used to estimate it based on water content.
- Q.25 Describe the steps involved in solving a Total Mass Balance problems for a fruit juice concentrator.
- Q.26 What is Enthalpy? How is it calculated for steam used in a food factory?
- Q.27 How do you calculate the energy required to heat 500kg of milk from 10°C to 75°C ? State the required formula.
- Q.28 Draw and explain the working principle of a U-Tube Manometer.

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